

1. Create an Incident Response Plan

Rubric Requirements:

- Detect at least 1 security incident
- Containment strategy
- Steps for eradication and recovery
- Identify and explain 1 type of cyber attack

How to demonstrate in your project:

- **Detection:** Use Wazuh alert rules to detect a specific attack, e.g., multiple failed SSH login attempts (brute force attack).
 - **Containment:** Show how alerts could trigger blocking the offending IP or isolating the affected system (you can describe this in the plan; actual network blocking can be simulated).
 - **Eradication & Recovery:** Explain steps like removing malware, resetting passwords, or restoring from backup. You can simulate this by creating a “malicious file” in Parrot and showing its detection and removal.
 - **Attack Type Example:** If using brute force, explain it as a **denial-of-service-style attack on login systems**; if simulating malware, show how Wazuh detects suspicious process executions or file changes.
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2. Develop a Comprehensive Security Policy

Rubric Requirements:

- At least 3 key security rules/guidelines
- Include incident response plan steps
- Explain how policies maintain **CIA Triad** (Confidentiality, Integrity, Availability)

How to demonstrate in your project:

- **Security Rules Example:**
 1. All login attempts must be logged and monitored (Confidentiality + Integrity)
 2. Critical system files must be protected and monitored for changes (Integrity)
 3. All security alerts must be reviewed within 24 hours (Availability + Integrity)
 - **Connection to Wazuh/Parrot:** Show Parrot OS logs collected by Wazuh, custom rules generating alerts, and how analysts would respond according to the policy.
 - **CIA Triad:** Explain how alert monitoring, log protection, and response policies preserve confidentiality, integrity, and availability of systems.
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3. Apply Encryption Techniques

Rubric Requirements:

- Show encrypted text and decrypted plaintext (AES example)
- Show text hashed using MD5 or SHA

How to demonstrate in your project:

- Use Python or terminal commands in Parrot OS:
 - Encrypt a sample text with AES, then decrypt it to show consistency.
 - Hash another sample using MD5 (or SHA1) and show the hash.
 - Document this in a report or a small Python script — even though Wazuh is not handling the encryption, this demonstrates understanding of cryptography as part of cybersecurity basics.
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4. Demonstrate Legal and Ethical Compliance

Rubric Requirements:

- Explain legal/ethical compliance in the incident response plan
- Identify 2 relevant laws/regulations
- Discuss at least 1 ethical consideration
- Explain how your plan upholds these

How to demonstrate in your project:

- **Laws/Regulations Example:** GDPR, HIPAA, or local cybersecurity laws for logging/monitoring
- **Ethical Consideration:** Monitoring user activity without violating privacy; responsible handling of alerts and sensitive logs
- **Connection to Wazuh:** Show that logs are collected securely, alerts are reviewed, and sensitive information is protected according to legal/ethical guidelines.